

Math 526, S06
Quiz 1

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (12 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 1 \end{bmatrix}$

a) Compute $\mathbf{AB}$ and $(\mathbf{AB})^t$.

Solution:

$$\mathbf{AB} = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -10 & -8 & 2 \\ -11 & -10 & 1 \\ -12 & -12 & 0 \end{bmatrix}$$

$$(\mathbf{AB})^t = \begin{bmatrix} -10 & -11 & -12 \\ -8 & -10 & -12 \\ 2 & 1 & 0 \end{bmatrix}$$

b) Compute $\mathbf{A}^t\mathbf{B}^t$. Does $(\mathbf{AB})^t = \mathbf{A}^t\mathbf{B}^t$?

Solution:

$$\mathbf{A}^t\mathbf{B}^t = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \begin{bmatrix} 1 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -4 & -4 & 0 \\ -7 & -10 & -3 \\ -10 & -16 & -6 \end{bmatrix}$$

So, we have

$$(\mathbf{AB})^t \neq \mathbf{A}^t\mathbf{B}^t$$

Problem 2. (8 points) Let $\mathbf{A} = \begin{bmatrix} -1 & 1 & -3 \\ 1 & -4 & 1 \\ -1 & -1 & -2 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 2 & -2 & 1 \\ 0 & -1 & 1 \end{bmatrix}$. Which side can $\mathbf{A}$ multiply $\mathbf{B}$ from? Explain the reason and compute the result.

Solution: Since $\mathbf{A}$ is 3×3 matrix and $\mathbf{B}$ is a 2×3 matrix, $\mathbf{A}$ only can multiply $\mathbf{B}$ from the right such that

$$\mathbf{BA} = \begin{bmatrix} 2 & -2 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & -3 \\ 1 & -4 & 1 \\ -1 & -1 & -2 \end{bmatrix} = \begin{bmatrix} -5 & 9 & -10 \\ -2 & 3 & -3 \end{bmatrix}$$